Earth's mini-moon: 2026 QQ8

CHEAT SHEET

@nova.xplains

What happened?

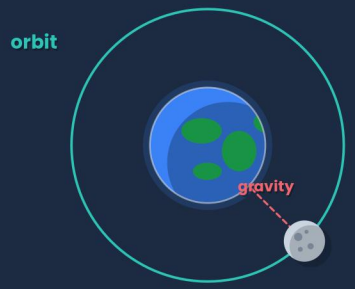


Asteroid 2026 QQ8 has been circling Earth since April 2026

Found Aug 20, 2026 by K. Sárneczky · confirmed natural Oct 29

Only about 1 to 2 m wide: as big as a fridge

What is a moon?

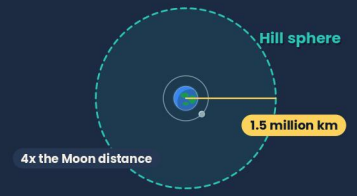


A moon orbits a planet, held by its gravity

Our Moon: 3,474 km wide, about 384,000 km away

Mini-moon: about 2 million times smaller, stays only months

The Hill sphere



Zone where Earth's gravity beats the Sun's

Reaches about 1.5 million km: about 4x the Moon distance

Outside it, the Sun's gravity wins

How Earth catches it



Most asteroids are too fast and just fly past

QQ8 orbits the Sun in 359 days, almost like Earth

It drifts in slowly, like a car next to you on the highway

Mini-moon vs quasi-moon



Mini-moon: truly captured, loops around Earth (months)

Quasi-moon: orbits the Sun next to Earth (e.g. 2025 PN7)

Quasi-moons can stay near us for decades

Why it leaves again



The Sun keeps tugging: wobbly loops, closest about 568,000 km

QQ8 leaves around the turn of the year 2026/27

It comes back about every 10 years · earlier: 2020 CD3, 2024 PT5

BONUS Quiz: Earth can only hold a mini-moon for months because the Sun's gravity pulls it away (answer B).

Sources: EarthSky (Nov 4, 2026); Space.com (Nov 5, 2026); Minor Planet Center; NASA/JPL (2020 CD3, 2024 PT5)

Earth's mini-moon: 2026 QQ8

CHEAT SHEET

@nova.xplains

What happened?

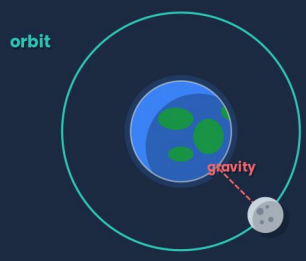


Asteroid 2026 QQ8 has been circling Earth since April 2026

Found Aug 20, 2026 by K. Sárneczky · confirmed natural Oct 29

Only about 1 to 2 m wide: as big as a fridge

What is a moon?

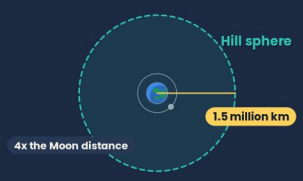


A moon orbits a planet, held by its gravity

Our Moon: 3,474 km wide, about 384,000 km away

Mini-moon: about 2 million times smaller, stays only months

The Hill sphere



Zone where Earth's gravity beats the Sun's

Reaches about 1.5 million km: about 4x the Moon distance

Outside it, the Sun's gravity wins

How Earth catches it



Most asteroids are too fast and just fly past

QQ8 orbits the Sun in 359 days, almost like Earth

It drifts in slowly, like a car next to you on the highway

Mini-moon vs quasi-moon

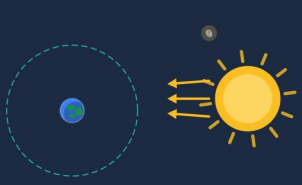


Mini-moon: truly captured, loops around Earth (months)

Quasi-moon: orbits the Sun next to Earth (e.g. 2025 PN7)

Quasi-moons can stay near us for decades

Why it leaves again



The Sun keeps tugging: wobbly loops, closest about 568,000 km

QQ8 leaves around the turn of the year 2026/27

It comes back about every 10 years · earlier: 2020 CD3, 2024 PT5

BONUS Quiz: Earth can only hold a mini-moon for months because the Sun's gravity pulls it away (answer B).

Sources: EarthSky (Nov 4, 2026); Space.com (Nov 5, 2026); Minor Planet Center; NASA/JPL (2020 CD3, 2024 PT5)